Methods Memo: How Does AI Improve Renewable Energy? A U.S. Test of Zhang et al. (2026)

Oct 22, 2026 · @AH

ECON 499R, ATL Economics (Prof. Caroline Fohlin). Team: Alexander Hendricks (finance side) and Shunji Lewandowski (modeling side). Memo due Friday, October 23, 2026. Last work day: Tuesday, November 24, 2026.

1. What we are testing

We will test whether AI (artificial intelligence, computer programs that learn from data and make decisions) helps the United States use less energy, get more output from each unit of energy, and make a bigger share of its electricity from renewable sources (energy that nature keeps refilling, like sunlight, wind, and flowing water). We copy the four hypotheses (educated guesses that data can confirm or reject) from Zhang, Ji, Dou, and Hao (2026), who studied 261 cities in China from 2006 to 2019 (p. 6503), and we run the same tests on the 50 U.S. states from 2015 to 2025.

Research question for this memo: Does AI improve renewable energy in the United States, and does it work by changing energy use, energy efficiency, and the energy mix, through a more tech-skilled workforce and denser economies, with spillovers (effects that leak across a border) into neighboring states?

Hypothesis	What the paper says (China)	What we will test (United States)
H1	AI lowers energy consumption, raises energy efficiency, and improves the energy structure (the mix of dirty and clean energy) (Zhang et al. 2026, p. 6498)	States with more AI activity use less electricity per person, earn more dollars of output per unit of electricity, and get a bigger share of their electricity from renewables
H2a	AI improves human-machine alignment (how well workers and AI tools fit together), and that indirectly helps the energy transition (p. 6498)	AI growth raises the share of tech workers in a state, and that tech workforce is part of why energy outcomes improve

Hypothesis	What the paper says (China)	What we will test (United States)
H2b	AI pulls in skilled workers and industries, creating economic agglomeration (lots of business activity packed into a small area), which indirectly drives the energy transition (p. 6498)	AI growth raises economic density (dollars of output per square mile), and that density is part of why energy outcomes improve
H3	AI has spatial spillover effects; because the siphoning effect (a hub pulling talent and money away from its neighbors) beats the trickle-down effect (a hub sharing gains with its neighbors), AI slows the energy transition in neighboring regions (p. 6499)	AI growth in one state makes the energy outcomes of its neighboring states worse, not better

What the paper found (p. 6509): (1) AI cut energy use, raised efficiency, and cleaned up the energy mix; (2) human-machine alignment and agglomeration were the two channels; (3) the effect was bigger where digital infrastructure, research spending, and labor-intensive industry were strong; (4) the effect grew as energy equity (a fair split of energy use and output across regions) improved; (5) neighbors were hurt because siphoning beat trickle-down. We test findings (1), (2), and (5) directly. Findings (3) and (4) are stretch goals (Section 5), because they need extra variables and a harder model.

One wording point: a regression (a math tool that measures how much one thing moves when another thing moves) can support or reject these hypotheses; it cannot prove them. The final report will say "support" or "reject."

2. The outcomes we study

We study the same three outcomes the paper used, rebuilt for U.S. states with free government data: energy consumption (how much), energy efficiency (how well), and energy structure (how clean) (Zhang et al. 2026, p. 6499).

Outcome	How the paper measured it (China)	How we will measure it (U.S. states)	Direction that means AI is helping
Energy consumption (EC)	A city's electricity, liquefied petroleum gas, and natural gas use, converted into one unit (p. 6499)	Megawatt-hours (MWh, a unit of electricity) sold in the state in a year, divided by the state's population. Electricity only, because data centers run on electricity	Lower
Energy efficiency (EE)	GDP (gross domestic product, the dollar value of everything produced in a year) divided by energy consumption (p. 6499)	State GDP in $ divided by MWh sold, so $ of output per MWh	Higher
Energy structure (ES), main version	Coal consumption divided by clean energy consumption; lower means cleaner (p. 6499)	Renewable share: MWh generated from solar, wind, hydro, geothermal, and biomass, divided by all MWh generated in the state	Higher
Energy structure (ES), paper-matching check	Same as above	Fossil-to-renewable ratio: MWh from coal, gas, and oil divided by renewable MWh; lower means cleaner, same direction as the paper	Lower
Renewable build-out (extra check for our question)	Not in the paper	New renewable capacity in megawatts (MW, the size of a plant) added in the state each year	Higher

Two cleaning steps copy the paper. First, we winsorize (pull the most extreme values in) at the 1st and 99th percentiles so one unusual state does not drive the answer (p. 6499). Second, we take the natural log (a math transform that turns big-versus-small gaps into percentage gaps) of consumption and efficiency, which is how the paper's consumption numbers look in its Table 1 (p. 6503).

We describe the outcomes before we test anything: a table of averages by year, and three U.S. maps (one per outcome, 2015 beside 2025), the way the paper shows China in its Figures 2 to 4 (p. 6500).

3. The cause we measure: AI

Our AI measure is data-center jobs per 1,000 workers in each state, each year. It plays the role the paper gave to industrial robots.

The paper could not see AI directly, so it used robot density (robots per worker) as a proxy (a stand-in you can count for something you cannot see), because robots are a physical, countable sign of AI in the real economy (Zhang et al. 2026, p. 6501). It took robot counts by industry from the IFR (International Federation of Robotics) and spread them across cities using each city's 2008 mix of industries (p. 6501, equation 1).

Data centers are the U.S. version of that idea. They are the buildings where AI models are trained and run, they are physical and countable, and the federal government counts their workers every year. We build the measure in three steps:

1. Take annual average employment in NAICS (North American Industry Classification System, the government's numbering of industries) code 518210, "Data Processing, Hosting, and Related Services," for each state, from the BLS (Bureau of Labor Statistics) QCEW (Quarterly Census of Employment and Wages).
2. Divide by total employment in the state (QCEW industry code 10, all industries).
3. Multiply by 1,000.

```
\boxed{AI_{st} = 1{,}000 \times \frac{\text{data-center jobs}_{st}}{\text{all jobs}_{st}}}
```

Where s is the state and t is the year. We do not need the paper's 2008 industry weights, because our numbers already come by state.

Why jobs and not megawatts: no free source gives data-center electricity capacity for every state for every year 2015 to 2025, but the QCEW jobs series is complete, free, and already verified for the project (project doc, data-coverage-matrix-2015-2026.md, Oct 9, 2026). Megawatts enter as a robustness check (Section 5) from the partial sources the project already found: Cleanview state pages, the Epoch AI data-center file, and the More Than Just Parks tracker.

4. Comparison and identification strategy

We compare each state with itself over time, and all states with each other within the same year. That is a two-way fixed effects design, the same one the paper used with city and year effects (Zhang et al. 2026, p. 6502, equation 4).

In plain words, fixed effects (constants that soak up everything that never changes) remove two kinds of noise. State fixed effects remove what is permanent about a state: its size, its weather, whether it has coal mines or big rivers. Year fixed effects remove what hits every state in the same year: national gas prices, federal tax credits, COVID-19 in 2020. What is left is the only thing we want: did a state's energy outcomes move in the years its AI measure moved, compared with other states in those same years?

Identification (the logic that lets us call a pattern a cause instead of a coincidence) is the hardest part, and we say so up front:

- Reverse causality (the effect causing the cause): data centers may choose states that already have cheap, plentiful, or clean power. Then renewables attract AI instead of AI improving renewables. The paper fixed this with an instrumental variable (a helper variable that moves AI but cannot move the outcome except through AI), using U.S. robot use as the helper for China (p. 6504, Table 3). We cannot copy that, because no outside country's data-center series moves with U.S. states and nothing else.
- Our fix, in order: (1) use last year's AI to explain this year's outcome (a one-year lag), so the outcome cannot have caused the cause; (2) control for the state's average electricity price, which is the main reason data centers pick a state; (3) stretch goal, a shift-share instrument (national data-center job growth multiplied by each state's 2015 share of data-center jobs), already written into the project's model plan (project doc, models-and-data-sources.md, Section 3, item 3), with the KP-LM (Kleibergen-Paap Lagrange Multiplier) test the paper reports to show the helper is strong enough (p. 6504).

- Standard errors (the uncertainty around each estimate) are clustered by state (allowed to be related within a state across years), so we do not overstate our confidence.

For H3 we also need to define a neighbor. The paper used a spatial weight matrix (a table that scores how close every pair of places is) built from the inverse of the distance between city centers (p. 6508). We do the same with state centroids (the center point of each state on a map), and we check the result with a border matrix (1 if two states share a border, 0 if not). Alaska and Hawaii have no land neighbors, so we run the spatial models with and without them.

Georgia is our home case, not a separate regression. Georgia Power's quarterly large-load reports to the Georgia PSC (Public Service Commission, the state board that approves utility plans), Docket 55378, give the data-center pipeline in megawatts from 2024 to 2026 (project doc, data-coverage-matrix-2015-2026.md). We use them for the interviews and for a one-page Georgia box in the final report.

5. Statistical methods, step by step

Four steps, run in this order, each a cut-down version of the paper's model. Steps 1 to 3 are committed for the Tuesday, November 24, 2026 draft; the stretch goals run only if Steps 1 to 3 are done by Tuesday, November 17.

Step 1. H1: two-way fixed effects panel regression

A panel (the same 50 states watched every year) regression with state and year fixed effects, copied from the paper's equation 4 (Zhang et al. 2026, p. 6502). We run it three times, once per outcome.

```
\boxed{Y_{st} = a_0 + a_1\, AI_{s,t-1} + a_2\, X_{st} + \mu_s + \tau_t +
\varepsilon_{st}}
```

Y is the outcome (energy consumption, efficiency, or renewable share) for state s in year t. AI is last year's data-center jobs per 1,000 workers. X is the list of controls (other things we hold steady so they do not confuse the answer): log GDP per person, manufacturing share of GDP, average electricity price, and population growth. The Greek letters are the state effect (mu), the year effect (tau), and the error (epsilon, everything we did not measure). H1 is supported if a1 is negative for consumption, positive for efficiency, and positive for renewable share, each with a p-value below 0.05 (a p-value is the chance of seeing a result this big by luck; below 5% counts as real).

Step 2. H2a and H2b: four-step mediation

A mediator (the middle step that carries an effect from the cause to the outcome) is tested with three more regressions, following the paper's equations 5 to 7 (p. 6502 to 6503). We do this twice: once with M = human-machine alignment (HMA) and once with M = economic agglomeration (EA).

$$\boxed{M_{st} = b_0 + b_1\, AI_{s,t-1} + b_2\, X_{st} + \mu_s + \tau_t + \varepsilon_{st}}$$

$$\boxed{Y_{st} = c_0 + c_1\, M_{st} + c_2\, X_{st} + \mu_s + \tau_t + \varepsilon_{st}}$$

$$\boxed{Y_{st} = e_0 + e_1\, M_{st} + e_2\, AI_{s,t-1} + e_3\, X_{st} + \mu_s + \tau_t + \varepsilon_{st}}$$

The paper's rule (p. 6503): if b1 is significantly positive (AI raises the mediator) and c1 and e1 are significant (the mediator moves the outcome), the channel is real. We also report the Sobel Z test (one number that checks whether the indirect path is real), as the paper does in its Tables 4 and 5 (p. 6505).

How we build the two mediators:

- HMA copies the paper's coupling formula (p. 6501, equation 2). AI is our measure above; hum is the share of the state's workers in computer systems design (NAICS 5415). Both are range-standardized first (rescaled to run from 0 to 1 using (value minus minimum) divided by (maximum minus minimum)).

$$\boxed{HMA_{st} = \sqrt{U_{st} \times V_{st}},\quad U_{st} = \frac{2\sqrt{AI_{st} \times hum_{st}}}{AI_{st} + hum_{st}},\quad V_{st} = 0.5\, AI_{st} + 0.5\, hum_{st}}$$

- EA copies the paper's economic density measure (p. 6502): non-farm GDP in $ divided by the state's land area in square miles.

$$\boxed{EA_{st} = \frac{\text{non-farm GDP}_{st}}{\text{land area}_s}}$$

Step 3. H3: neighbors and spillovers

First, Moran's I (a score from minus 1 to plus 1 that says whether states near each other have similar values), one per outcome per year, using the paper's equation 10 (p. 6508). A positive, significant score means the outcomes cluster in space, so a spatial model is needed.

```
\boxed{I = \frac{n}{\sum_{i}\sum_{j} w_{ij}} \cdot \frac{\sum_{i}\sum_{j}
w_{ij}\, z_i z_j}{\sum_{i} z_i^2}}
```

n is the number of states, w is the weight between state i and state j (bigger when closer), and z is each state's distance from the average.

Second, the SDM (Spatial Durbin Model, a regression that also includes the neighbors' outcome and the neighbors' AI), copied from the paper's equation 11 (p. 6508).

```
\boxed{Y_{st} = \rho \sum_{j} W_{sj} Y_{jt} + \beta_1\, AI_{s,t-1} +
\beta_2\, X_{st} + \theta_1 \sum_{j} W_{sj} AI_{j,t-1} + \theta_2 \sum_{j}
W_{sj} X_{jt} + \mu_s + \tau_t + \varepsilon_{st}}
```

Theta-1 is the neighbor effect of AI. H3 (siphoning) is supported if theta-1 has the harmful sign: positive for consumption, negative for efficiency, negative for renewable share. That matches the paper's Table 12, where the neighbor AI term raised consumption, lowered efficiency, and worsened the energy mix (p. 6509). Fallback if the SDM will not run by November 17: the SLX model (spatial lag of X), which only adds the neighbors' average AI to the Step 1 equation and runs in the same software. It tests the same theta-1.

Step 4. Robustness checks

Robustness checks (redoing the math a few different ways to see if the answer holds) follow the paper's Appendix B list (p. 6505): replace the outcome (renewable share with the fossil-to-renewable ratio); replace the cause (data-center jobs with data-center megawatts where available); change the sample (drop 2020, drop Alaska and Hawaii, drop Virginia, the largest data-center state); and re-run with clustered standard errors.

Stretch goals (only if Steps 1 to 3 are done by November 17)

- Moderation (does the AI effect change with a state's conditions?): add interaction terms (AI multiplied by broadband subscription rate, and AI multiplied by research spending share), as in the paper's equation 8 (p. 6503).

- Threshold (does the effect switch on only above some level?): Hansen (1999) threshold model with energy equity, the ratio of a state's GDP share to its electricity share, as in the paper's equation 3 (p. 6502).
- Instrumental variable: the shift-share instrument from Section 4 with the KP-LM test.

Software and who does what

R with the fixest package (Steps 1, 2, 4 and the SLX fallback) and the spdep and spatialreg packages (Moran's I and SDM); Stata (xtreg, fe and xsmle) is the alternative. Shunji Lewandowski runs the models. Alexander Hendricks builds and checks every variable in a spreadsheet before it enters R, and writes the Data and Results text.

6. Data needed and sources

Every core variable comes from a free federal file that was opened and verified either in the project on October 9, 2026 (project doc, data-coverage-matrix-2015-2026.md) or today, October 22, 2026, so the core panel of 50 states from 2015 to 2025 is feasible. Three EIA state summary files found today (sales, generation, and capacity by state) replace most of the plant-level aggregation the project first planned.

Variable (role)	Paper's version (page)	Our U.S. version	Source and years	Status
Energy consumption (outcome)	City electricity, LPG, and gas use (p. 6499)	MWh sold per person	EIA (U.S. Energy Information Administration) state file HS861 2010-.xlsx, annual sales to customers by state and sector, 2010 to 2025 (released Oct 7, 2026); Census county population 2020 to 2025 and BEA population 1969 to 2024; backup: EIA-861 utility files 2015 to 2025	State file verified Oct 22, 2026; backup verified in project
Energy consumption, paper-matching check	Same (p. 6499)	All energy use (electricity plus every other fuel), per person	EIA SEDS (State Energy Data System), 1960 to 2024 (released Jun 26, 2026; 2025 arrives June 2027)	Verified Oct 22, 2026

Variable (role)	Paper's version (page)	Our U.S. version	Source and years	Status
Energy efficiency (outcome)	GDP divided by energy (p. 6499)	State GDP in $ divided by MWh	BEA (Bureau of Economic Analysis) state GDP tables, quarterly series through Q2 2026 (released Sep 30, 2026; next Dec 23, 2026), so 2025 is the four 2025 quarters; backup: county GDP CAGDP1, 2001 to 2024, summed to states (2025 county data arrive Dec 2, 2026)	BEA page verified Oct 22, 2026; state table to confirm on first download
Energy structure (outcome)	Coal divided by clean energy (p. 6499)	Renewable share of MWh generated; fossil-to-renewable ratio	EIA state file annual_generation_state.xlsx, net generation by state, producer type, and energy source, 1990 to 2025 (released Sep 14, 2026); backup: EIA-923 plant files 2015 to 2025	State file verified Oct 22, 2026; backup verified in project
Renewable build-out (check)	Not in the paper	Renewable MW in service each year; new MW is the change from the year before	EIA state file existcapacity_annual.xlsx, capacity by energy source, producer type, and state, 1990 to 2025 (released Sep 10, 2026); backup: EIA-860 generator files	State file verified Oct 22, 2026; backup verified in project
AI (cause)	Robot density from IFR data (p. 6501)	Data-center jobs per 1,000 workers	BLS (Bureau of Labor Statistics) QCEW (Quarterly Census of Employment and Wages), NAICS 518210, annual 2015 to 2025 and 2026 Q1; QCEW industry code 10 for all jobs, same download pattern, column annual_avg_emplvl, state total rows coded agglvl_code 50	518210 verified in project; code 10 file to confirm on first download (BLS blocks automated checks)
hum, tech-worker share (for HMA)	Share of workers in information, computer services, and software (p. 6501)	Share of workers in NAICS 5415, computer systems design	BLS QCEW, NAICS 5415, annual 2015 to 2025, same download pattern, state industry rows coded agglvl_code 56	To confirm on first download (BLS blocks automated checks)

Variable (role)	Paper's version (page)	Our U.S. version	Source and years	Status
Economic agglomeration (mediator)	Non-farm value added divided by area (p. 6502)	Non-farm GDP in $ divided by land area in square miles	BEA GDP by industry (state tables, or county CAGDP2 2001 to 2024); Census 2010 state area file (Georgia: 57,513 square miles)	GDP verified in project; land area verified Oct 22, 2026
Controls	ln GDP per person, government size, industry mix, transport, finance (p. 6502)	ln GDP per person (BEA); manufacturing share of GDP (BEA); average electricity price, revenue divided by sales (EIA HS861 file); population growth (Census)	Same files as above	Verified
Spatial weights (H3)	Inverse distance between cities (p. 6508)	Inverse distance between state centroids; shared-border matrix as a check	Census TIGER state boundary file	To verify when Step 3 starts
Data-center MW (robustness)	Not in the paper	Operating MW by state	Cleanview state pages, Epoch AI data-center CSV, More Than Just Parks tracker (all Oct 2026)	Verified in project; partial coverage
Georgia case	Not in the paper	Georgia Power large-load pipeline, MW by quarter	Georgia PSC Docket 55378, Q2 2024 to Q2 2026	Verified in project
Stretch: broadband share (moderator)	Broadband users divided by population (p. 6502)	Households with a broadband subscription, %	Census ACS (American Community Survey) table S2801 through the Census API (free key); 2024 variable S2801_C02_014E, "Broadband of any type," percent; variable codes change by year; 1-year tables confirmed for 2015 and 2024	API page verified Oct 22, 2026; state rows to confirm on first query

Variable (role)	Paper's version (page)	Our U.S. version	Source and years	Status
Stretch: research spending share (moderator)	Science spending divided by fiscal spending (p. 6502)	R&D (research and development) spending divided by GDP	NSF (National Science Foundation) state R&D tables	Not yet checked
Stretch: weather control	Not in the paper	Heating and cooling degree days (days cold enough to heat or hot enough to cool, base 65°F)	NOAA (National Oceanic and Atmospheric Administration) Climate at a Glance statewide series, annual heating and monthly cooling degree days, CSV download	Verified Oct 22, 2026 (Georgia 2015 to 2025 tested)
Stretch: renewable mandate control	Not in the paper	State RPS (renewable portfolio standard, a law requiring a share of clean power) target by year	LBNL (Lawrence Berkeley National Laboratory) spreadsheet "RPS & CES Targets and Demand," June 2026, from the 2026 Data Update	Verified Oct 22, 2026
Stretch: second AI measure	Robot density	AI job postings as a share of all postings, by state	Stanford AI Index Report 2026, Chapter 4 (Economy), Figure 4.4.13, from Lightcast data, 2010 to 2025; public data folder linked from the report	Verified Oct 22, 2026; selected states only, so it cannot replace QCEW for all 50

Sample size: 50 states over 11 years gives 550 state-years for all three outcomes (2015 to 2025) if the BEA state GDP table for 2025 checks out on download; otherwise efficiency stops at 2024 with 500. The paper had 3,654 city-years (p. 6503). Every annual file in the core list is under 25 MB (project doc, data-coverage-matrix-2015-2026.md), so the downloads take minutes, and the panel build is about two weeks of part-time work (Section 8).

7. Interviews

We will do two 30-minute interviews, by phone or Zoom, between Monday, October 26 and Friday, November 13, 2026. Requests go out Friday, October 23. Interview answers are quotes and examples for the report; they do not go into the regressions.

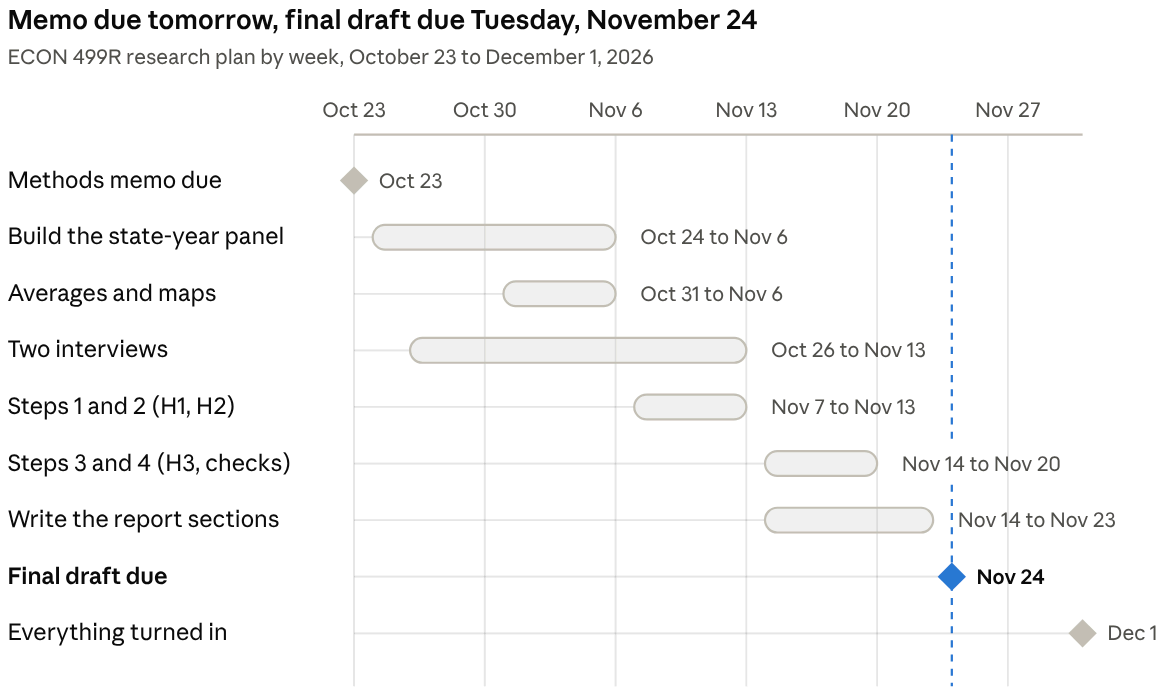
Who (pick two of three)	Why this person	Hypotheses it speaks to
A staff member at the Georgia PSC, or a resource planner at Georgia Power	They file and review the data-center pipeline numbers in Docket 55378 and decide what mix of gas, solar, and batteries Georgia builds for that load	H1 (energy mix), H3 (does Georgia's data-center wave pull from or push to neighboring states)
An energy or sustainability manager at an Atlanta-area data-center operator	They decide whether a site buys renewable power, runs AI tools to cut its own power use, and hires tech-skilled workers	H1 (efficiency), H2a (worker and machine fit), H2b (why they cluster near Atlanta)
An energy economist at Emory or Georgia Tech	They can tell us which controls matter in the United States and whether our one-year lag is enough for the reverse-causality problem	Identification (Section 4)

Five questions for every interview, so answers can be compared:

1. In the last five years, how has data-center demand changed what your organization builds or buys: more gas, more solar and wind, more batteries, or no change?
2. Has AI (in your own operations or in your customers') cut the electricity needed per unit of work? Can you give one example with a number?
3. Do the people who run these systems need different skills than five years ago? Which skills?
4. Why do data centers cluster around Atlanta instead of spreading out across Georgia and neighboring states?
5. If a neighboring state wins a big data-center campus, does Georgia gain, lose, or neither? Why?

Before the first interview, we confirm with Prof. Fohlin whether the course requires any approval or consent form for interviewing professionals about their work.

8. Timeline



research timeline · 9 items, final draft November 24

The panel build and the interviews run first, the regressions take the two weeks after, the final draft is due on the last work day, Tuesday, November 24, 2026, and everything is turned in by Tuesday, December 1. At the course pace of 6 hours a week per person, that is about 30 hours each from the memo to the final draft.

9. Limitations we expect

The biggest limit is that we cannot copy the paper's instrumental variable, so our results show association (two things moving together) with a one-year lag, not airtight cause and effect.

1. Reverse causality. Data centers may go where power is already cheap and clean. The lag and the electricity-price control reduce this; the shift-share instrument is a stretch goal, not a promise (Section 4).

2. The AI measure counts jobs, not computing. A data center can use a great deal of electricity while employing only a small staff, so jobs can understate AI in some states. The megawatt check in Step 4 is partial, because no free source covers every state every year (project doc, data-coverage-matrix-2015-2026.md, atlas-gap findings).
3. Small sample. The paper had 261 cities over 14 years, 3,654 observations (Zhang et al. 2026, p. 6503). We have 50 states over 10 to 11 years, about 500 to 550 observations. Fewer observations means less statistical power (the ability to spot a real effect), so a true effect could show up as "not significant."
4. Electricity crosses state lines. Our renewable share counts what is generated inside a state, but a state can import clean power from a neighbor. The spatial model in Step 3 partly handles this.
5. Missing and unusual years. County GDP for 2025 is not published until December 2, 2026, so efficiency for 2025 depends on BEA's state GDP table (quarterly through Q2 2026), which we confirm on download; 2020 is distorted by COVID-19 and is dropped in a check; 2026 is incomplete and left out.
6. Time. Mediation and spatial models are new to both of us. If the panel build runs late, Step 3 falls back to the simpler SLX model and the stretch goals are dropped.
7. Interviews are two people. Their answers are examples, not evidence that a pattern holds everywhere.
8. Policy that changes within a state. State fixed effects remove only fixed policy. A state that raises its renewable mandate mid-sample could look like an AI effect. A renewable-mandate control is a stretch goal pending data.
9. Georgia is regulated. Georgia Power and the Georgia PSC decide the energy mix, not a market, so Georgia's numbers may not follow the market logic the paper assumes. We treat Georgia as a case box, not as the test.

10. What we expect to find

We expect the United States to match the paper on efficiency, the two channels, and spillovers, but to flip the sign on consumption, because our AI measure counts data centers and data centers eat electricity.

Hypothesis	Paper's result for China	What we expect for the U.S.	Why
H1, consumption	AI lowered energy use (Zhang et al. 2026, Table 2, p. 6504)	Opposite sign: more data-center jobs, more electricity per person	The paper's own review warns that AI "drives energy consumption through data centre scale effects" (p. 6497). Our proxy is that scale effect
H1, efficiency	AI raised GDP per unit of energy (Table 2, p. 6504)	Same sign, positive	Data-center states tend to be high-output service economies, and fixed effects will test whether the gain shows up within a state as its AI grows
H1, energy mix	AI improved the mix (Table 2, p. 6504)	Positive on renewable share in market states; weak or zero in regulated states like Georgia	Georgia Power's response to its data-center pipeline (Docket 55378) is decided by the utility and the Georgia PSC, not by a market, so the paper's market logic may not apply there (Section 9)
H2a, worker and machine fit	Channel confirmed (Table 4, p. 6505)	Same, positive	Data-center growth and computer-services jobs grow in the same states
H2b, agglomeration	Channel confirmed (Table 5, p. 6505)	Same, positive	Data centers cluster in dense metro areas such as Atlanta

Hypothesis	Paper's result for China	What we expect for the U.S.	Why
H3, neighbors	Siphoning: neighbors got worse (Table 12, p. 6509)	Siphoning likely, trickle-down possible	A hub state can pull projects from its neighbors, but when a hub fills up the next wave lands next door; our interview question 5 asks Georgia planners which one they see

If the consumption sign flips as expected, the honest headline is that the paper's H1 holds in the United States for efficiency and the energy mix but not for total use, which is itself a result worth reporting.

11. Sources

The only literature source in this memo is the paper being transposed; body citations use author-date with page numbers, and the Works Cited entry below is in MLA (swap the DOI for the EBSCOhost Emory proxy link in the final report).

Zhang, Shengling, Ruibing Ji, Wei Dou, and Yu Hao. "Blessing or Peril? The Impact of Artificial Intelligence on the Energy Transition: Evidence from China." Applied Economics, vol. 58, no. 32, 2026, pp. 6495 to 6513. https://doi.org/10.1080/00036846.2025.2521038.

Project documents (ECON 499R project, both dated October 9, 2026): models-and-data-sources.md (the earlier model plan, including the shift-share instrument) and data-coverage-matrix-2015-2026.md (every data link below was opened and checked there, with file sizes and status codes).

Data files, latest vintage, as verified in the project doc (earlier years follow the same address pattern in the archive folder):

- EIA-860, 2025 final (23.6 MB) and EIA-860M, August 2026 (14.0 MB)
- EIA-923, 2025 final (23.3 MB)
- EIA-861, 2025 final (4.8 MB)
- BLS QCEW, NAICS 518210, 2025 annual (0.3 MB)

- BEA CAGDP1 (1.9 MB), BEA CAGDP2 (15.3 MB), BEA CAINC1 (3.5 MB); BEA county release page (next release Dec 2, 2026)
- Census county population, vintage 2025 (2.1 MB)
- Georgia PSC Docket 55378, Georgia Power large-load reports, Q2 2024 to Q2 2026
- Data-center megawatts, partial: Cleanview U.S. map, Epoch AI data centers, More Than Just Parks tracker

Data pages opened and checked on October 22, 2026, for this memo:

- EIA state electricity data page: annual_generation_state.xlsx (1990 to 2025, released Sep 14, 2026), existcapacity_annual.xlsx (1990 to 2025, released Sep 10, 2026), HS861 2010-.xlsx (sales by state and sector, 2010 to 2025, released Oct 7, 2026)
- EIA SEDS complete data: 1960 to 2024, released Jun 26, 2026
- BEA GDP by state page: Q2 2026 release dated Sep 30, 2026; next release Dec 23, 2026
- Census 2010 state area measurements: land area in square miles by state
- Census API variable page, ACS 2024 S2801_C02_014E: percent of households with broadband of any type
- NOAA Climate at a Glance, statewide time series; Georgia heating-degree-day test file: 2015 to 2025 CSV
- LBNL Renewables Portfolio Standards resources: "U.S. State Electricity Resource Standards: 2026 Data Update" and the spreadsheet "RPS & CES Targets and Demand" (June 2026)
- Stanford AI Index Report 2026, Chapter 4, Economy: Figure 4.4.13, AI share of job postings by state
- BLS QCEW files for NAICS 5415 and for all industries (code 10) use the same address pattern as the verified 518210 file, with 5415 or 10 in place of 518210; BLS blocks automated checks, so open each in a browser before the first download